

Abhishek Shukla

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Summary

- Software Engineer with 3+ years of experience building scalable backend systems, search platforms, and AI-powered agentic workflows in production environments.
- Strong expertise in Python, FastAPI, Elasticsearch, and AWS, with hands-on experience in multi-agent systems and LLM-driven applications.
- Experienced in designing and delivering end-user facing production systems, collaborating with stakeholders, and driving solutions from requirements to deployment.

Experience

Cactus Communications

Mar 2023 – Present

Software Engineer

Remote

- **Designed and owned backend systems** powering large-scale search and AI-driven workflows, handling **200M+ searchable records** and **1M+ API requests**.
- **Architected and optimized Elasticsearch infrastructure**, including index redesign, schema optimization, and sharding strategies for high-performance search at scale.
- **Monitored and debugged production systems** using Grafana and Kibana, identifying bottlenecks and improving system latency and reliability.
- **Designed and built production-grade multi-agent systems** using AutoGen, enabling complex workflow orchestration across LLM-powered agents, enabling context-aware reasoning, state management, and improved response quality across interactions
- **Integrated LLMs into production APIs**, enabling intelligent automation, summarization, and decision-making workflows.
- **Built LLM-based evaluation systems**, using models as evaluators and decision engines to validate outputs and improve system accuracy.
- **Debugged and optimized agent workflows in production**, identifying failure modes (hallucinations, latency, cost spikes) and improving robustness and reliability.
- **Applied prompt engineering and structured prompting techniques** to ensure deterministic and high-quality agent outputs.
- **Benchmarked LLMs and agent configurations**, evaluating performance, latency, and cost trade-offs across different models and workflows.
- **Reduced processing time by 80%** and operational cost by **65%** through optimization of agent pipelines and infrastructure usage.
- **Collaborated with DevOps teams** to scale services on AWS (ECS, EC2, Lambda, SQS, DynamoDB), ensuring high availability and resilience.
- **Owned end-to-end lifecycle** from system design to deployment, monitoring, and continuous optimization of backend and AI systems.

Cactus Communications

Sep 2022 – Feb 2023

Software Engineer Intern

Remote

- Designed and implemented a **real-time notification system** using AWS Lambda and Slack integration.
- Built **SQS-triggered pipelines** for asynchronous processing and automated S3 backups.
- Improved reliability of microservices by resolving issues across QA, staging, and production environments.

Dotvik Solutions

Mar 2022 – Jun 2022

Full Stack Developer Intern

Noida, India

- Redesigned order APIs, improving processing efficiency by **90%**.
- Optimized email templating system using Velocity engine.

Project

Ask R Discovery

AI Research Assistant

- Built an AI-powered research assistant used by researchers to answer complex queries with citation-backed responses.
- Designed a **multi-agent workflow** for intent detection, document retrieval, and LLM-based summarization over large-scale research data.

Skills

- **Languages:** Python, Java
- **Backend:** FastAPI, Flask, Spring Boot, Microservices
- **AI/ML:** LLMs, Agentic Systems, Prompt Engineering, AutoGen
- **Search/Data:** Elasticsearch, DynamoDB
- **Cloud:** AWS (Lambda, ECS, EC2, S3, SQS, CloudWatch)
- **Tools:** Docker, Jenkins, Git, Kibana, Grafana

Education

Poornima University

Aug 2019 – May 2023

B.Tech in Computer Science (CGPA: 8.6)

Jaipur, Rajasthan